Sample MERN with Microservices

For `helloService`, create `.env` file with the content:

```bash
PORT=3001
```

For `profileService`, create `.env` file with the content:

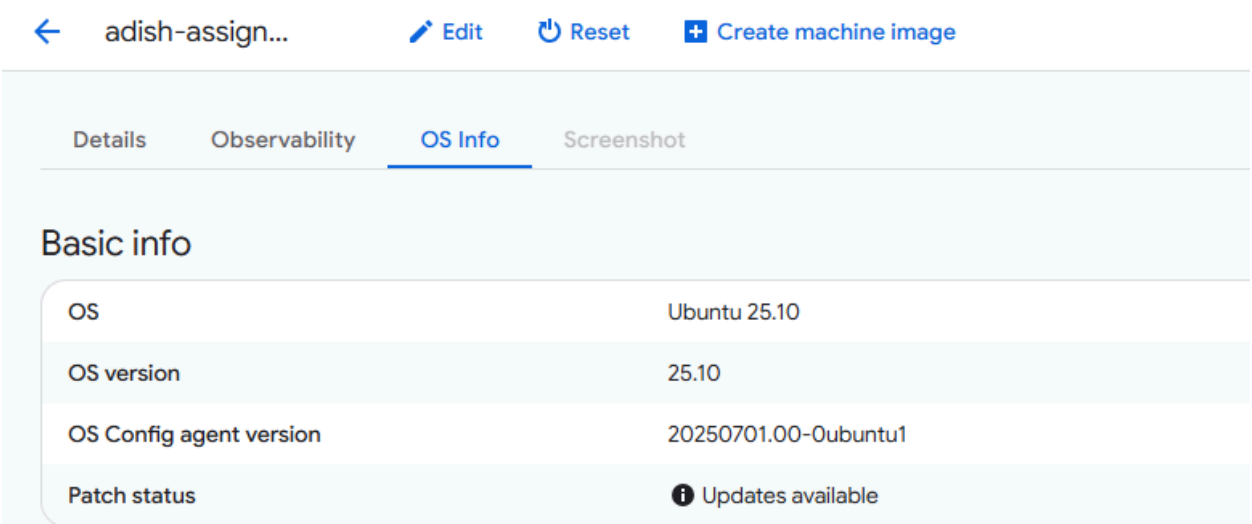
```bash
PORT=3002
MONGO_URL="specifyYourMongoURLHereWithDatabaseNameInTheEnd"
```

Finally install packages in both the services by running the command `npm install`.

For frontend, you have to install and start the frontend server:

```bash
cd frontend
npm install
npm start
```

Note: This will run the frontend in the development server. To run in production, build the application by running the command `npm run build



Install and upate the apt package

sudo apt update
sudo apt upgrade -y
sudo apt install git -y
git clone https://github.com/Adish786/SampleMERNwithMicroservices.git
cd SampleMERNwithMicroservices
sudo mkdir k8s-manifests

Install Docker
In simple ways only run the single command
sudo apt install docker.io -y

Or If you want to install with certificate and add the repository then follow the next step

```
# Add Docker's official GPG key
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor
-o /usr/share/keyrings/docker-archive-keyring.gpg

# Add the Docker repository
echo "deb [arch=$(dpkg --print-architecture)
signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null

# Install Docker Engine
sudo apt update
sudo apt install -y docker-ce docker-ce-cli containerd.io
```

```
# Add your user to the 'docker' group to run Docker commands without 'sudo'
sudo usermod -aG docker $USER
```

#Check the systemctl status

sudo systemctl status containerd

systemctl enable --now containerd

systemctl restart docker

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# sudo usermod -aG docker $USER
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# systemctl status docker
× docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: failed (Result: exit-code) since Tue 2026-04-14 08:42:04 UTC; 17s ago
   Duration: 48min 54.782s
 Invocation: f64380eb7a9a4cc3901aa459acce07c0
TriggeredBy: × docker.socket
       Docs: https://docs.docker.com
    Process: 10711 ExecStart=/usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock (code=exited, status=1/FAILURE)
   Main PID: 10711 (code=exited, status=1/FAILURE)

Apr 14 08:42:04 adish-assignment systemd[1]: docker.service: Scheduled restart job, restart counter is at 3.
Apr 14 08:42:04 adish-assignment systemd[1]: docker.service: Start request repeated too quickly.
Apr 14 08:42:04 adish-assignment systemd[1]: docker.service: Failed with result 'exit-code'.
Apr 14 08:42:04 adish-assignment systemd[1]: Failed to start docker.service - Docker Application Container Engine.
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# ^C
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# systemctl enable --now containerd
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# systemctl status containerd
● containerd.service - containerd container runtime
     Loaded: loaded (/usr/lib/systemd/system/containerd.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-04-14 08:41:56 UTC; 3min 22s ago
 Invocation: 4aef35a7204b4ec3bcfe643672f52dec
       Docs: https://containerd.io
   Main PID: 10535 (containerd)
      Tasks: 7
     Memory: 13.8M (peak: 13.8M)
        CPU: 392ms
     CGroup: /system.slice/containerd.service
             └─10535 /usr/bin/containerd

Apr 14 08:41:56 adish-assignment containerd[10535]: time="2026-04-14T08:41:56.501910373Z" level=info msg="loading plugin" id=io.containerd.monitor.container.v
```

Install kubectl

Download the latest release
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"

# Install the binary
sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl

Verify the installation
kubectl version --client

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/am
d64/kubectl"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 55.8M  100 55.8M    0     0   132M      0 --:--:-- --:--:-- --:--:--  132M
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl version --client
Client Version: v1.35.3
Kustomize Version: v5.7.1
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices#
```

Install Minikube

Download the latest Minikube binary
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64

# Install it
sudo install minikube-linux-amd64 /usr/local/bin/minikube

Verify the installation
minikube version

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/am
d64/kubectl"
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 55.8M  100 55.8M    0     0   132M      0 --:--:-- --:--:-- --:--:--  132M
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl version --client
Client Version: v1.35.3
Kustomize Version: v5.7.1
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  128M  100  128M    0     0   154M      0 --:--:-- --:--:-- --:--:--  154M
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# sudo install minikube-linux-amd64 /usr/local/bin/minikube
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices#
```

minikube start --driver=docker

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  128M  100  128M    0     0   154M      0 --:--:-- --:--:-- --:--:--  154M
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# sudo install minikube-linux-amd64 /usr/local/bin/minikube
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# minikube start --driver=docker
😄 minikube v1.38.1 on Ubuntu 25.10 (amd64)
✨ Using the docker driver based on user configuration

💣 Exiting due to PROVIDER_DOCKER_NOT_RUNNING: "docker version --format <no value>-<no value>:<no value>" exit status 1: Cannot connect to the Docker daemon at unix:///var
/run/docker.sock. Is the docker daemon running?
💡 Suggestion: Start the Docker service
📘 Documentation: https://minikube.sigs.k8s.io/docs/drivers/docker/

root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices#
```

Install Google Cloud SDK

Download and install gcloud CLI
curl -O
https://dl.google.com/dl/cloudsdk/channels/rapid/downloads/google-cloud-cli-linux-x86_64.tar.gz
tar -xf google-cloud-cli-linux-x86_64.tar.gz
./google-cloud-sdk/install.sh
Restart your shell or run:
source ~/.bashrc

Authenticate and Set Project

gcloud auth login
gcloud config set project YOUR_PROJECT_ID # replace with your GCP project ID

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud auth login

You are running on a Google Compute Engine virtual machine.
It is recommended that you use service accounts for authentication.

You can run:

  $ gcloud config set account `ACCOUNT`

to switch accounts if necessary.

Your credentials may be visible to others with access to this
virtual machine. Are you sure you want to authenticate with
your personal account?

Do you want to continue (Y/n)?  y

Go to the following link in your browser, and complete the sign-in prompts:

    https://accounts.google.com/o/oauth2/auth?response_type=code&client_id=32555940559.apps.googleusercontent.com&redirect_uri=https%3A%2F%2Fsdk.cloud.google.com%2Fauthcod
e.html&scope=openid+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fuserinfo.email+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcloud-platform+https%3A%2F%2Fwww.googleapis.com%2Fau
th%2Fappengine.admin+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fsqlservice.login+https%3A%2F%2Fwww.googleapis.com%2Fauth%2Fcompute+https%3A%2F%2Fwww.googleapis.com%2Fauth%2
Faccounts.reauth&state=Dq8oGaNZrlR9ZCcNJYHdCbwi6RseoA&prompt=consent&token_usage=remote&access_type=offline&code_challenge=JhY7yox1PnEjOIrE4gZJLfXLjsQw2ylQmYTY5952-HQ&code
_challenge_method=S256

Once finished, enter the verification code provided in your browser: 4/0Aci98E9QKmoasRPwJcaVcs3UQ3rIbfEA4bvbTupCSlEmjwd2b2J1SQHCXMDU9Gv32TeIMw

You are now logged in as [adishansari786@gmail.com].
Your current project is [heroviredacademics].  You can change this setting by running:
  $ gcloud config set project PROJECT_ID
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud config set project heroviredacademics
[environment: untagged] Read more to tag: g.co/cloud/project-env-tag.
Updated property [core/project].
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices#
```

Create a Docker file for each service

helloservice

```
# Build stage
FROM node:18-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm ci --only=production

# Runtime stage
FROM node:18-alpine
```

```
RUN addgroup -g 1001 -S nodejs && adduser -S nodejs -u 1001
WORKDIR /app
COPY --from=builder --chown=nodejs:nodejs /app/node_modules ./node_modules
COPY . .
USER nodejs
EXPOSE 3001
CMD ["node", "index.js"]
```

profileservice

```
# Build stage
FROM node:18-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm ci --only=production

# Runtime stage
FROM node:18-alpine
RUN addgroup -g 1001 -S nodejs && adduser -S nodejs -u 1001
WORKDIR /app
COPY --from=builder --chown=nodejs:nodejs /app/node_modules ./node_modules
COPY . .
USER nodejs
EXPOSE 3002
CMD ["node", "index.js"]
```

frontendservice

```
# Stage 1: Build the React application
FROM node:18-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm ci
COPY . .
RUN npm run build

# Stage 2: Serve with Nginx
FROM nginx:stable-alpine
# Copy custom nginx config (optional - see note below)
COPY nginx.conf /etc/nginx/conf.d/default.conf
```

```
# Copy built static files
COPY --from=builder /app/build /usr/share/nginx/html
EXPOSE 3000
CMD ["nginx", "-g", "daemon off;"]
```

In frontendservice

`nginx.conf`

```
cat > nginx.conf << 'EOF'
server {
    listen 3000;
    server_name localhost;
    root /usr/share/nginx/html;
    index index.html;

    location / {
        try_files $uri $uri/ /index.html;
    }

    location ~* \.(js|css|png|jpg|jpeg|gif|ico|svg)$ {
        expires 1y;
        add_header Cache-Control "public, immutable";
    }
}
EOF
```

Build & Push Images Using Cloud Build

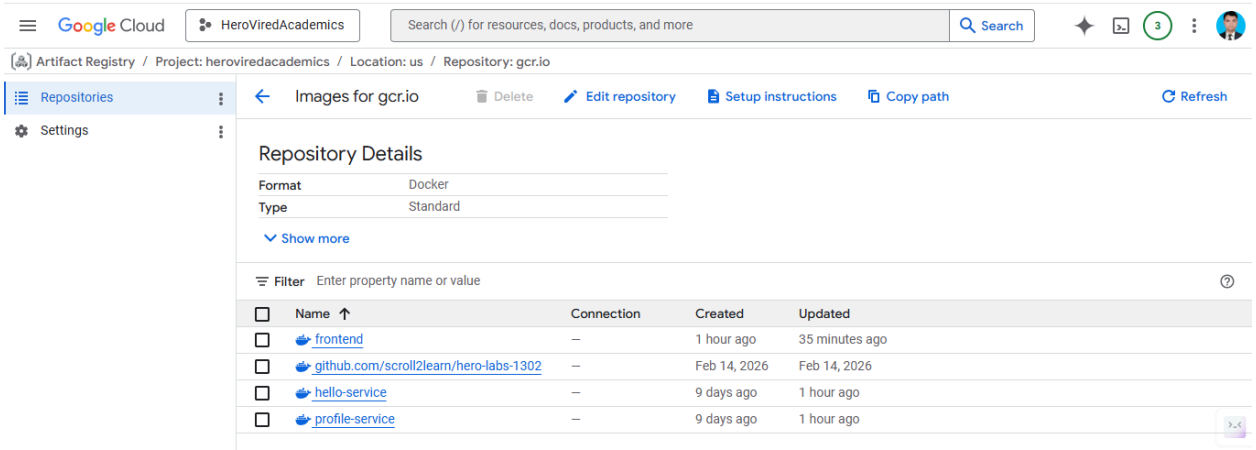
Enable Cloud Build API

gcloud services enable cloudbuild.googleapis.com
gcloud builds submit --tag gcr.io/heroviredacademics/hello-service ./backend/helloService
gcloud builds submit --tag gcr.io/heroviredacademics/profile-service ./backend/profileService
gcloud builds submit --tag gcr.io/heroviredacademics/frontend ./frontend

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud builds submit --tag gcr.io/heroviredacademics/hello-service ./backend/helloService
Creating temporary archive of 4 file(s) totalling 38.5 KiB before compression.
Uploading tarball of [./backend/helloService] to [gs://heroviredacademics_cloudbuild/source/1776159392.287754-3b621200039b452fba4f552ed3643117.tgz]
Created [https://cloudbuild.googleapis.com/v1/projects/heroviredacademics/locations/global/builds/29c867af-816b-4aa3-9fa6-077c064712d9].
Logs are available at [ https://console.cloud.google.com/cloud-build/builds/29c867af-816b-4aa3-9fa6-077c064712d9?project=601575894555 ].
Waiting for build to complete. Polling interval: 1 second(s).
------------------------------------------------------------------------- REMOTE BUILD OUTPUT -------------------------------------------------------------------------
starting build "29c867af-816b-4aa3-9fa6-077c064712d9"

FETCHSOURCE
Fetching storage object: gs://heroviredacademics_cloudbuild/source/1776159392.287754-3b621200039b452fba4f552ed3643117.tgz#1776159392554844
Copying gs://heroviredacademics_cloudbuild/source/1776159392.287754-3b621200039b452fba4f552ed3643117.tgz#1776159392554844...
/ [1 files][ 11.8 KiB/ 11.8 KiB]
Operation completed over 1 objects/11.8 KiB.
BUILD
Already have image (with digest): gcr.io/cloud-builders/gcb-internal
Sending build context to Docker daemon  43.01kB
Step 1/12 : FROM node:18-alpine AS builder
18-alpine: Pulling from library/node
f18232174bc9: Pulling fs layer
dd71dde834b5: Pulling fs layer
1e5a4c89cee5: Pulling fs layer
25ff2da83641: Pulling fs layer
25ff2da83641: Verifying Checksum
25ff2da83641: Download complete
1e5a4c89cee5: Verifying Checksum
1e5a4c89cee5: Download complete
f18232174bc9: Verifying Checksum
f18232174bc9: Download complete
f18232174bc9: Pull complete
dd71dde834b5: Verifying Checksum
dd71dde834b5: Download complete
dd71dde834b5: Pull complete
1e5a4c89cee5: Pull complete
25ff2da83641: Pull complete
Digest: sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
Status: Downloaded newer image for node:18-alpine
 ---> ee77c6cd7c18
Step 2/12 : WORKDIR /app
 ---> Running in c4e550aaf08e
```

```
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
ID                                      CREATE_TIME                DURATION  SOURCE
  IMAGES                                                           STATUS
29c867af-816b-4aa3-9fa6-077c064712d9  2026-04-14T09:36:32+00:00  25S       gs://heroviredacademics_cloudbuild/source/1776159392.287754-3b621200039b452fba4f552ed3643117.tgz
  gcr.io/heroviredacademics/hello-service (+1 more)  SUCCESS
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud builds submit --tag gcr.io/heroviredacademics/profile-service ./backend/profileService
Creating temporary archive of 4 file(s) totalling 48.4 KiB before compression.
Uploading tarball of [./backend/profileService] to [gs://heroviredacademics_cloudbuild/source/1776159435.251703-e354016306aa4b72aa483019dd1150e6.tgz]
Created [https://cloudbuild.googleapis.com/v1/projects/heroviredacademics/locations/global/builds/987b75fc-3016-47d9-bb71-9b110c7e55a3].
Logs are available at [ https://console.cloud.google.com/cloud-build/builds/987b75fc-3016-47d9-bb71-9b110c7e55a3?project=601575894555 ].
Waiting for build to complete. Polling interval: 1 second(s).
------------------------------------------------------------------------- REMOTE BUILD OUTPUT -------------------------------------------------------------------------
starting build "987b75fc-3016-47d9-bb71-9b110c7e55a3"

FETCHSOURCE
Fetching storage object: gs://heroviredacademics_cloudbuild/source/1776159435.251703-e354016306aa4b72aa483019dd1150e6.tgz#1776159435457698
Copying gs://heroviredacademics_cloudbuild/source/1776159435.251703-e354016306aa4b72aa483019dd1150e6.tgz#1776159435457698...
/ [1 files][ 14.5 KiB/ 14.5 KiB]
Operation completed over 1 objects/14.5 KiB.
BUILD
Already have image (with digest): gcr.io/cloud-builders/gcb-internal
Sending build context to Docker daemon  53.25kB
Step 1/12 : FROM node:18-alpine AS builder
18-alpine: Pulling from library/node
f18232174bc9: Pulling fs layer
dd71dde834b5: Pulling fs layer
1e5a4c89cee5: Pulling fs layer
25ff2da83641: Pulling fs layer
1e5a4c89cee5: Verifying Checksum
```

```
-----------------------------------------------------------------------------------------------------------------------------------------------------------------
ID                                      CREATE_TIME                DURATION  SOURCE
  IMAGES                                                           STATUS
987b75fc-3016-47d9-bb71-9b110c7e55a3  2026-04-14T09:37:15+00:00  26S       gs://heroviredacademics_cloudbuild/source/1776159435.251703-e354016306aa4b72aa483019dd1150e6.tg
  gcr.io/heroviredacademics/profile-service (+1 more)  SUCCESS
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud builds submit --tag gcr.io/heroviredacademics/frontend ./frontend
Creating temporary archive of 19 file(s) totalling 717.7 KiB before compression.
Some files were not included in the source upload.

Check the gcloud log [/root/.config/gcloud/logs/2026.04.14/09.39.26.043734.log] to see which files and the contents of the
default gcloudignore file used (see `$ gcloud topic gcloudignore` to learn
more).

Uploading tarball of [./frontend] to [gs://heroviredacademics_cloudbuild/source/1776159566.295144-c17091e23225496e872c1fe7330dc606.tgz]
Created [https://cloudbuild.googleapis.com/v1/projects/heroviredacademics/locations/global/builds/33f37018-0ef2-4a6d-971e-acc3cf1bccc3].
Logs are available at [ https://console.cloud.google.com/cloud-build/builds/33f37018-0ef2-4a6d-971e-acc3cf1bccc3?project=601575894555 ].
Waiting for build to complete. Polling interval: 1 second(s).
------------------------------------------------------------------------- REMOTE BUILD OUTPUT -------------------------------------------------------------------------
starting build "33f37018-0ef2-4a6d-971e-acc3cf1bccc3"

FETCHSOURCE
Fetching storage object: gs://heroviredacademics_cloudbuild/source/1776159566.295144-c17091e23225496e872c1fe7330dc606.tgz#1776159566622605
Copying gs://heroviredacademics_cloudbuild/source/1776159566.295144-c17091e23225496e872c1fe7330dc606.tgz#1776159566622605...
/ [1 files][176.3 KiB/176.3 KiB]
Operation completed over 1 objects/176.3 KiB.
BUILD
Already have image (with digest): gcr.io/cloud-builders/gcb-internal
Sending build context to Docker daemon  752.1kB
Step 1/11 : FROM node:18-alpine AS builder
18-alpine: Pulling from library/node
f18232174bc9: Pulling fs layer
dd71dde834b5: Pulling fs layer
1e5a4c89cee5: Pulling fs layer
25ff2da83641: Pulling fs layer
25ff2da83641: Verifying Checksum
25ff2da83641: Download complete
f18232174bc9: Verifying Checksum
f18232174bc9: Download complete
1e5a4c89cee5: Verifying Checksum
1e5a4c89cee5: Download complete
```



gcloud container images list --repository=gcr.io/heroviredacademics

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud container images list --repository=gcr.io/heroviredacademics
NAME
gcr.io/heroviredacademics/frontend
gcr.io/heroviredacademics/github.com
gcr.io/heroviredacademics/hello-service
gcr.io/heroviredacademics/profile-service
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices#
```

Install the GKE Auth Plugin

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud container clusters create mern-cluster --zone us-central1-b --num-nodes 2  --machine-type e2
-medium
Note: Your Pod address range (`--cluster-ipv4-cidr`) can accommodate at most 1008 node(s).
Creating cluster mern-cluster in us-central1-b... Cluster is being health-checked (Kubernetes Control Plane is healthy)...done.
Created [https://container.googleapis.com/v1/projects/heroviredacademics/zones/us-central1-b/clusters/mern-cluster].
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload_/gcloud/us-central1-b/mern-cluster?project=heroviredacademics
CRITICAL: ACTION REQUIRED: gke-gcloud-auth-plugin, which is needed for continued use of kubectl, was not found or is not executable. Install gke-gcloud-auth-plugin for use
 with kubectl by following https://cloud.google.com/kubernetes-engine/docs/how-to/cluster-access-for-kubectl#install_plugin
kubeconfig entry generated for mern-cluster.
NAME          LOCATION       MASTER_VERSION      MASTER_IP        MACHINE_TYPE  NODE_VERSION        NUM_NODES  STATUS   STACK_TYPE
mern-cluster  us-central1-b  1.35.1-gke.1396002  104.154.231.234  e2-medium     1.35.1-gke.1396002  2          RUNNING  IPV4
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# ^C
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# sudo apt-get install google-cloud-sdk-gke-gcloud-auth-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-cloud-sdk-gke-gcloud-auth-plugin
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gke-gcloud-auth-plugin --version
bash: gke-gcloud-auth-plugin: command not found
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud components install gke-gcloud-auth-plugin


Your current Google Cloud CLI version is: 564.0.0
Installing components from version: 564.0.0
```

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud components install gke-gcloud-auth-plugin


Your current Google Cloud CLI version is: 564.0.0
Installing components from version: 564.0.0

┌──────────────────────────────────────────────────────────────────────────────┐
│                              These components will be installed.                              │
├──────────────────────────────────────────────────────────┬────────────┬──────────┤
│                                   Name                                   │   Version  │   Size   │
├──────────────────────────────────────────────────────────┼────────────┼──────────┤
│                gke-gcloud-auth-plugin (Platform Specific)                │   0.5.12   │ 3.8 MiB  │
└──────────────────────────────────────────────────────────┴────────────┴──────────┘

For the latest full release notes, please visit:
  https://cloud.google.com/sdk/release_notes

Once started, canceling this operation may leave your SDK installation in an inconsistent state.

Do you want to continue (Y/n)?  y

Performing in place update...

╠═ Downloading: gke-gcloud-auth-plugin                                        ═╣
╠═ Downloading: gke-gcloud-auth-plugin (Platform Specific)                    ═╣
╠═ Installing: gke-gcloud-auth-plugin                                         ═╣
╠═ Installing: gke-gcloud-auth-plugin (Platform Specific)                     ═╣

Performing post processing steps...done.

Update done!
```

```
Performing post processing steps...done.

Update done!

WARNING:   There are other instances of Google Cloud tools on your system PATH.
  Please remove the following to avoid confusion or accidental invocation:

  /usr/bin/snap


root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gke-gcloud-auth-plugin --version
Kubernetes v1.34.2+0dd7f7cd0b632699e47ecafa4acc8f77cfc73c06
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# gcloud container clusters get-credentials mern-cluster --zone us-central1-b
Fetching cluster endpoint and auth data.
kubeconfig entry generated for mern-cluster.
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices#
```

# Install the plugin using apt (Ubuntu)
sudo apt-get install google-cloud-sdk-gke-gcloud-auth-plugin //if apt are not found the plugin then run the second command

gcloud components install gke-gcloud-auth-plugin

Verify installation
gke-gcloud-auth-plugin --version

Create a GKE Cluster

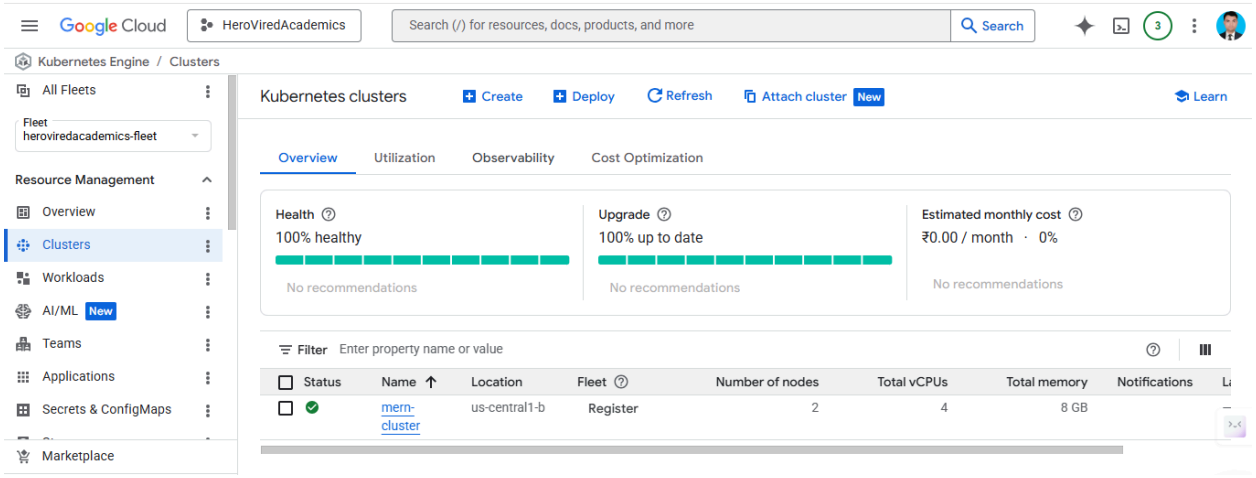
gcloud container clusters create mern-cluster --zone us-central1-b --num-nodes 2 --machine-type e2-medium

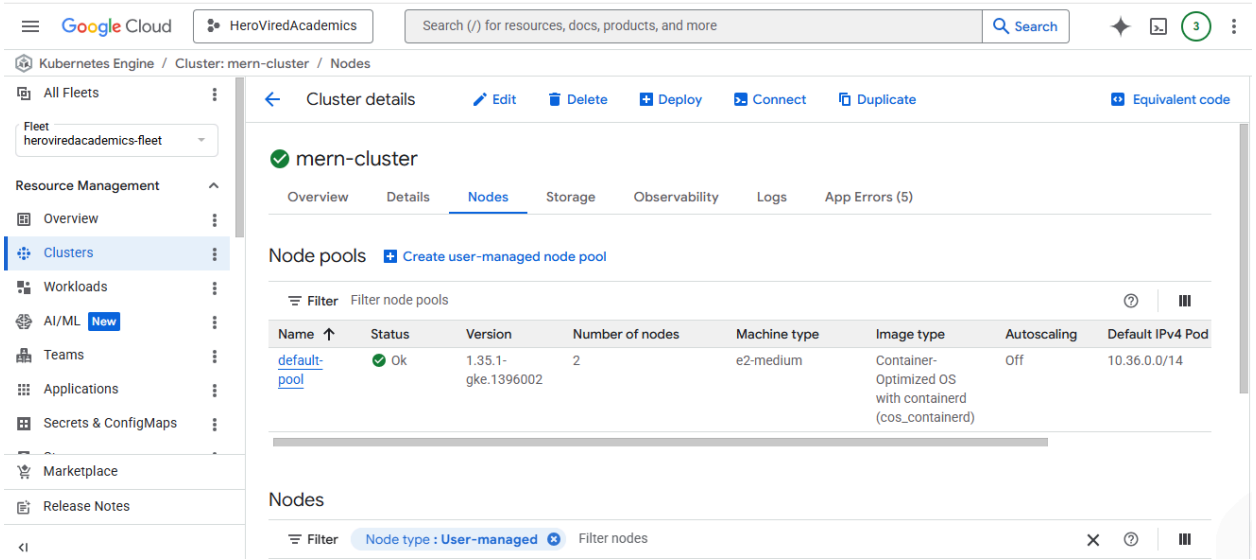
Add a Third Node

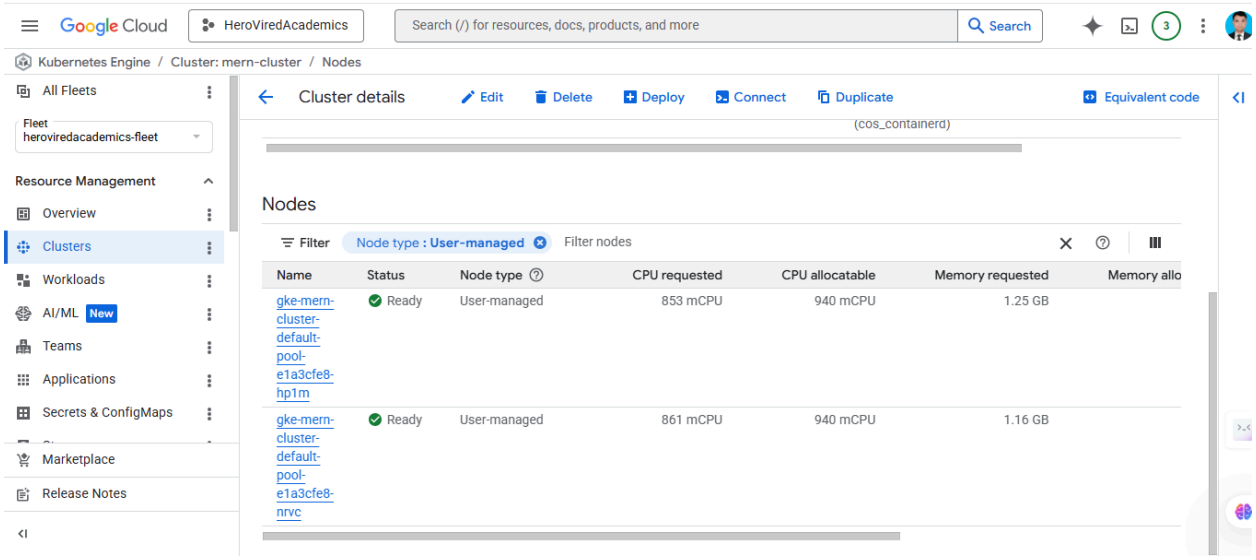
gcloud container clusters resize mern-cluster --node-pool default-pool --num-nodes=3 --zone us-central1-b

Get Credentials for kubectl

gcloud container clusters get-credentials mern-cluster --zone us-central1-b







Verify kubectl Works

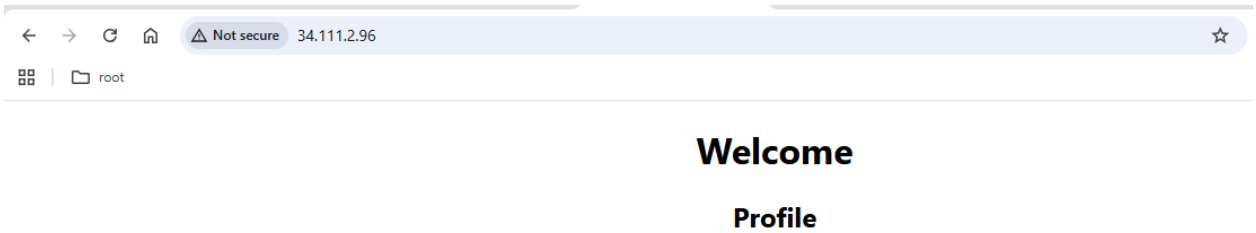
kubectl apply -f k8s-manifests/

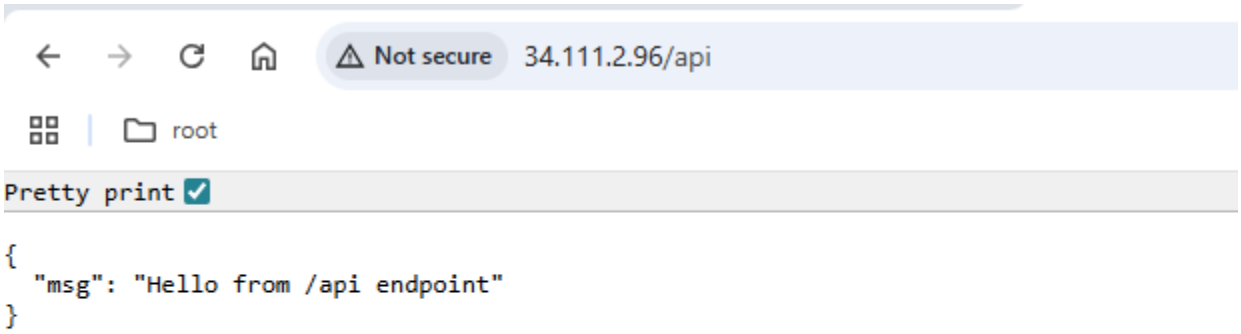
kubectl get nodes
kubectl get all
kubectl get pods
kubectl get pods -w
kubectl get svc
kubectl get ingress
kubectl get ingress mern-ingress //name of the cluster

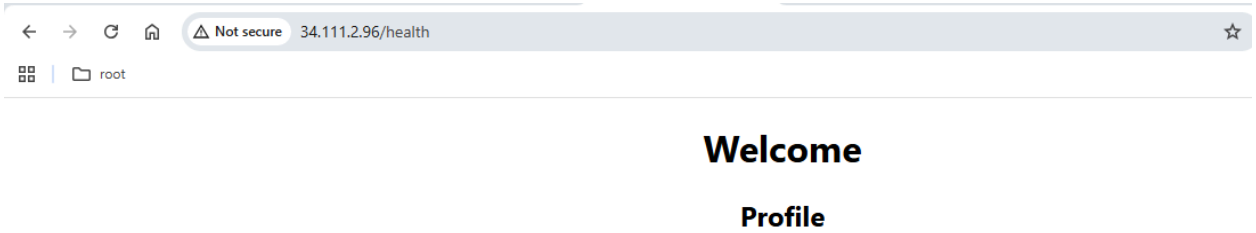
```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
frontend-68c74cbff7-4pjtj           1/1     Running   0          42h
frontend-68c74cbff7-jsf7m           1/1     Running   0          42h
hello-service-6d468b7878-j8szn      1/1     Running   0          41h
hello-service-6d468b7878-llnct      1/1     Running   0          41h
mongodb-0                           1/1     Running   0          42h
profile-service-5d87858c87-95trs    1/1     Running   0          56m
profile-service-5d87858c87-cpbbk    1/1     Running   0          56m
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl get nodes
NAME                                          STATUS   ROLES    AGE   VERSION
gke-mern-cluster-default-pool-e1a3cfe8-ggww   Ready    <none>   41h   v1.35.1-gke.1396002
gke-mern-cluster-default-pool-e1a3cfe8-hp1m   Ready    <none>   42h   v1.35.1-gke.1396002
gke-mern-cluster-default-pool-e1a3cfe8-nrvc   Ready    <none>   42h   v1.35.1-gke.1396002
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl get svc
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)     AGE
frontend-service   ClusterIP   34.118.235.30   <none>        3000/TCP    42h
hello-service      ClusterIP   34.118.227.71   <none>        3001/TCP    42h
kubernetes         ClusterIP   34.118.224.1    <none>        443/TCP     42h
mongodb-service    ClusterIP   None            <none>        27017/TCP   42h
profile-service    ClusterIP   34.118.235.36   <none>        3002/TCP    42h
```

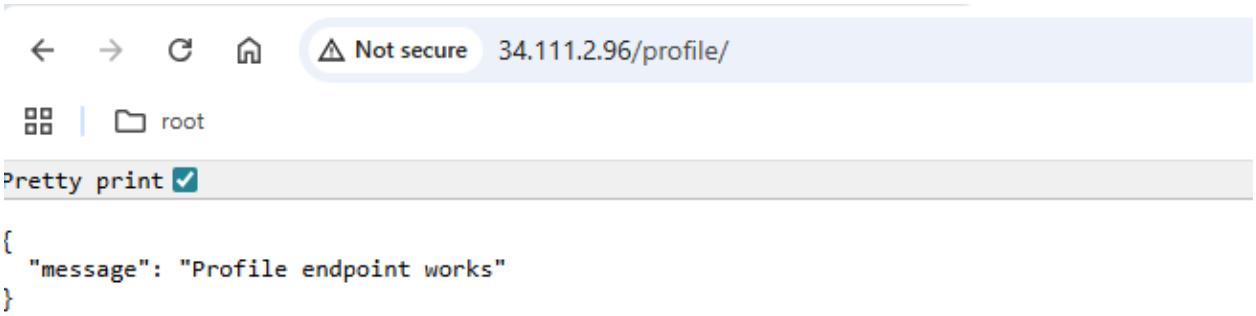
```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl get all
NAME                                 READY   STATUS    RESTARTS   AGE
pod/frontend-68c74cbff7-4pjtj        1/1     Running   0          42h
pod/frontend-68c74cbff7-jsf7m        1/1     Running   0          42h
pod/hello-service-6d468b7878-j8szn   1/1     Running   0          41h
pod/hello-service-6d468b7878-llnct   1/1     Running   0          41h
pod/mongodb-0                        1/1     Running   0          42h
pod/profile-service-5d87858c87-95trs 1/1     Running   0          56m
pod/profile-service-5d87858c87-cpbbk 1/1     Running   0          56m

NAME                       TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)     AGE
service/frontend-service   ClusterIP   34.118.235.30   <none>        3000/TCP    42h
service/hello-service      ClusterIP   34.118.227.71   <none>        3001/TCP    42h
service/kubernetes         ClusterIP   34.118.224.1    <none>        443/TCP     42h
service/mongodb-service    ClusterIP   None            <none>        27017/TCP   42h
service/profile-service    ClusterIP   34.118.235.36   <none>        3002/TCP    42h

NAME                              READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/frontend          2/2     2            2           42h
deployment.apps/hello-service     2/2     2            2           42h
deployment.apps/profile-service   2/2     2            2           42h

NAME                                         DESIRED   CURRENT   READY   AGE
replicaset.apps/frontend-68c74cbff7          2         2         2       42h
replicaset.apps/frontend-799bc57f64          0         0         0       42h
replicaset.apps/hello-service-68667b4f9b     0         0         0       42h
replicaset.apps/hello-service-6d468b7878     2         2         2       41h
replicaset.apps/profile-service-59b58b46cb   0         0         0       42h
replicaset.apps/profile-service-59c89f69d4   0         0         0       41h
replicaset.apps/profile-service-5d87858c87   2         2         2       62m
replicaset.apps/profile-service-647444457c   0         0         0       42h
replicaset.apps/profile-service-6bd9868446   0         0         0       25h
replicaset.apps/profile-service-6d77c54cf4   0         0         0       25h
replicaset.apps/profile-service-7448d88678   0         0         0       25h
replicaset.apps/profile-service-7575789dd7   0         0         0       25h
replicaset.apps/profile-service-7666cbf95d   0         0         0       25h
replicaset.apps/profile-service-767f579f45   0         0         0       41h
replicaset.apps/profile-service-c68ff5fcc    0         0         0       25h
```

```
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl get ingress mern-ingress
NAME           CLASS    HOSTS   ADDRESS       PORTS   AGE
mern-ingress   <none>   *       34.111.2.96   80      49m
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# kubectl get ingress
NAME           CLASS    HOSTS   ADDRESS       PORTS   AGE
mern-ingress   <none>   *       34.111.2.96   80      49m
root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# curl http://34.111.2.96
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="/favicon.ico"/><meta name="viewport" content="width=device-width,initial-scale=1"/><meta
 name="theme-color" content="#000000"/><meta name="description" content="Web site created using create-react-app"/><link rel="apple-touch-icon" href="/logo192.png"/><link
rel="manifest" href="/manifest.json"/><title>React App</title><script defer="defer" src="/static/js/main.7c7a6051.js"></script><link href="/static/css/main.f855e6bc.css" r
el="stylesheet"></head><body><noscript>You need to enable JavaScript to run this app.</noscript><div id="root"></div></body></html>root@adish-assignment:/home/adishansari7
86/SampleMERNwithMicroservices# curl http://34.111.2.96/api
{"msg":"Hello from /api endpoint"}root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices# curl http://34.111.2.96/health
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="icon" href="/favicon.ico"/><meta name="viewport" content="width=device-width,initial-scale=1"/><meta
 name="theme-color" content="#000000"/><meta name="description" content="Web site created using create-react-app"/><link rel="apple-touch-icon" href="/logo192.png"/><link
rel="manifest" href="/manifest.json"/><title>React App</title><script defer="defer" src="/static/js/main.7c7a6051.js"></script><link href="/static/css/main.f855e6bc.css" r
el="stylesheet"></head><body><noscript>You need to enable JavaScript to run this app.</noscript><div id="root"></div></body></html>root@adish-assignment:/home/adishansari7
86/SampleMERNwithMicroservices# curl http://34.111.2.96/profile
{"message":"Profile endpoint works"}root@adish-assignment:/home/adishansari786/SampleMERNwithMicroservices#
```









Frontend (should return the React HTML)
curl http://34.49.202.235/
Hello service (should return a greeting JSON)
curl http://34.49.202.235/api/
Profile service (should return profile data or a message)
curl http://34.49.202.235/profile/